

Multiple Choice

1. **Answer:** B

Options: A) Finding a longest simple cycle in G; B) Finding a shortest cycle in G; C) Finding ALL spanning trees of G; D) Finding a largest clique in G

Note: Correct option: B - Finding a shortest cycle in G

2. **Answer:** A

Options: A) `int(x [,base])`; B) `long(x [,base] )`; C) `float(x)`; D) `str(x)`

Note: Correct option: A - `int(x [,base])`

3. **Answer:** B

Options: A) Error; B) aab; C) ab; D) a ab

Note: Correct option: B - aab

4. **Answer:** C

Options: A) Error; B) abc; C) cba; D) c

Note: Correct option: C - cba

5. **Answer:** C

Options: A) [19,21]; B) [11,15]; C) [11,15,19]; D) [13,17,21]

Note: Correct option: C - [11,15,19]

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def extract_symmetric(test_list): | return (res)`
Note: Students should provide a detailed explanation referencing algorithm steps.
12. **Answer:** `def zip_tuples(test_tup 1, test_tup 2): | return (res)`
Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key